TEAM SAFEPATH

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SAFEPATH AI



ARTIFICIAL INTELLIGENCE

AI route prediction for safer, faster logistics

PREDICT. AVOID. DELIVER.

Business Analysis 3.2 (AIBUY3A) | Capstone Project
Vaal University of Technology | 02 November 2026

AI SOLUTION & BACKGROUND

THE BACKGROUND

- Delays come from accidents, traffic, weather and road closures
- Navigation apps pick routes by distance or travel time, so drivers meet obstacles nobody warned them about
- Local carriers face long delivery times and parcels stolen from trucks
- The result: higher delivery time, fuel use and operating cost

THE AI SOLUTION

- SafePath is an AI model that predicts delays and unsafe conditions on a route and redirects drivers before problems arise
- Scan, predict, redirect: checks candidate routes, forecasts traffic and ETA, then steers to the safest, fastest path
- Built on supervised machine learning: a tuned Random Forest and Decision Tree
- It advises the driver. It does not drive the truck.

BUSINESS OBJECTIVES

STRATEGIC OBJECTIVES

- Minimize travel time
- Enhance safety on every route
- Reduce fuel and maintenance costs
- Improve customer satisfaction
- Drive decisions with real-time data
- Boost business efficiency
- SafeMap AI being used in everyday commercial vehicles

HOW WE MEASURE SUCCESS

- Traffic state accuracy 99.8% on unseen days
- Volume forecast error 17.8 vehicles per 15 min (R^2 0.84)
- Driver questions understood 100% of the time on 22 unseen phrases
- A closed road is never recommended

DELIVERY DELAYS START ON THE ROAD

Navigation apps pick routes by distance or travel time, so drivers meet obstacles nobody warned them about.

ROAD ACCIDENTS
Sudden blockages and added risk

HEAVY TRAFFIC
Bottlenecks that burn time and fuel

SEVERE WEATHER
Slower, riskier driving conditions

ROAD CLOSURES
Unexpected detours mid-delivery

WHAT IT COSTS

- Longer delivery times
- More fuel burned idling
- Higher operating costs
- Parcels stolen from trucks

CURRENT NAVIGATION WORKS IN SILOS

TODAY'S NAVIGATION APPS

- Optimize for shortest distance or historical travel time
- Treat accidents, closures, potholes, theft and weather as static alerts
- Send drivers into bottlenecks and hazard zones
- Leave fleets with delays, higher fuel costs and more vehicle damage

SAFEPATH

- Puts safety parameters inside the core route calculation
- Turns accidents, closures, potholes, theft risk and weather into path-blocking metrics
- Cross-references all conditions at once
- Predicts the safest and most efficient path before problems arise

SAFEPATH: SCAN, PREDICT, REDIRECT

1

SCAN

Checks candidate routes for accidents, traffic, weather, road closures and other hazards.

2

PREDICT

Forecasts traffic levels and estimated travel time for each route before the truck leaves.

3

REDIRECT

Steers the driver to the safest and most efficient route before problems arise.



Why we built it

SafePath grew out of delivery problems in our local communities: long delivery times and parcels stolen from trucks. The model does not drive the truck. It helps drivers avoid traffic and safety risks, saving time and fuel.

WHAT THE SYSTEM MUST DO



CAR ACCIDENTS

Ingest real-time collision alerts, calculate delay around active zones, trigger alternative routes and warn the driver.



TRAFFIC RATE

Stream vehicle flow and segment speeds, update the ETA and re-check the route every 30 seconds.



POTHOLES

Build an infrastructure severity index from municipal reports and penalize roads with many potholes.



VEHICLE THEFT

Sync crime data and filter out high-theft sectors, especially at night and on long hauls.

UNIFIED PROCESSING

A cloud-hosted supervised regression model scores all four risk variables together and shows a single 'safest + shortest' recommendation on a low-distraction driver dashboard.

CONSTRAINTS AND RISKS TO MANAGE

CONSTRAINTS

Dependence on external APIs
Mapping, traffic, pothole and theft data all come from third parties such as Google Maps, HERE and municipal databases.

Constant connectivity
Live accident and traffic streams need steady 4G/5G. Dead zones force cached data.

High compute and battery use
Re-scoring 3 to 5 routes every 30 seconds loads servers and drains devices.

RISKS

API outages
Provider downtime blinds the system mid-trip

Cellular dead zones
Lost signal freezes the 30-second rerouting

Peak-hour server load
Scanning 3 to 5 routes per vehicle strains the cloud

Secondary jams
Sending every fleet down one 'safe' road creates a new jam

Privacy and skewed data
Live tracking raises privacy risk; unbalanced data distorts risk

IMPLEMENTATION TOOLS



PYTHON AND SCIKIT-LEARN

Build, train, tune and validate the models



PANDAS AND NUMPY

Data cleaning, lag features and feature engineering



GOOGLE MAPS / MAPBOX

Planned: real road geometry and distances (routes are simulated for now)



TOMTOM / HERE

Planned: live traffic speeds and accident feeds



OPENWEATHERMAP

Planned: live weather to replace the weather severity score



FIREBASE / AWS

Planned: cloud hosting for deployment (the prototype runs on Flask)

MACHINE LEARNING APPROACH

Supervised learning on past data with known outcomes. Two tasks, each compared with a simple baseline and then tuned.

TASK 1: REGRESSION

Predict how many cars will be on the road in the next 15 minutes

Tested 4 models: Linear Regression, Random Forest, Gradient Boosting, Neural Network (MLP)

17.80

vehicles mean error per 15 min, tuned Random Forest

TASK 2: CLASSIFICATION

Show traffic as low, normal, high or heavy on the map

Tested 3 models: Logistic Regression, Decision Tree, Random Forest

99.8%

correct on unseen days, tuned Decision Tree

DATA & DATASET

	A	B	C	D	E	F	G	H	I	J	K	L
1	Time	Date	Day of the week	CarCount	BikeCount	BusCount	TruckCount	Total	Traffic Situation	Weather	Accidents	Road Closed
2	12:00:00 am	10	Tuesday	13	2	2	24	41	normal	Light snow	0	No
3	12:15:00 am	10	Tuesday	14	1	1	36	52	normal	Light snow	0	No
4	12:30:00 am	10	Tuesday	10	2	2	32	46	normal	Moderate	0	No
5	12:45:00 am	10	Tuesday	10	2	2	36	50	normal	Moderate	0	No
6	1:00:00 am	10	Tuesday	11	2	1	34	48	normal	Moderate	0	No
7	1:15:00 am	10	Tuesday	15	1	1	39	56	normal	Moderate	0	No
8	1:30:00 am	10	Tuesday	14	2	2	27	45	normal	Moderate	0	No
9	1:45:00 am	10	Tuesday	13	2	1	20	36	normal	Moderate	0	No
10	2:00:00 am	10	Tuesday	7	0	0	26	33	normal	Light snow	0	No

Traffic class	Records	Share of data
Normal	**3,610**	**60.7%**
Heavy	**1,137**	**19.1%**
Low	**834**	**14.0%**
High	**371**	**6.2%**
Total	**5,952**	**100%**

Normal dominates and 'high' is rare, so class weights balance training.

ABOUT THE DATASET

- 5,952 records over 62 days at 15-minute intervals
- Inputs: vehicle counts, weather, accidents, road closures, time and day, and past traffic patterns
- Split by day: 4,608 rows to train, 1,248 to test
- Class weights offset the imbalance; the test days stay untouched

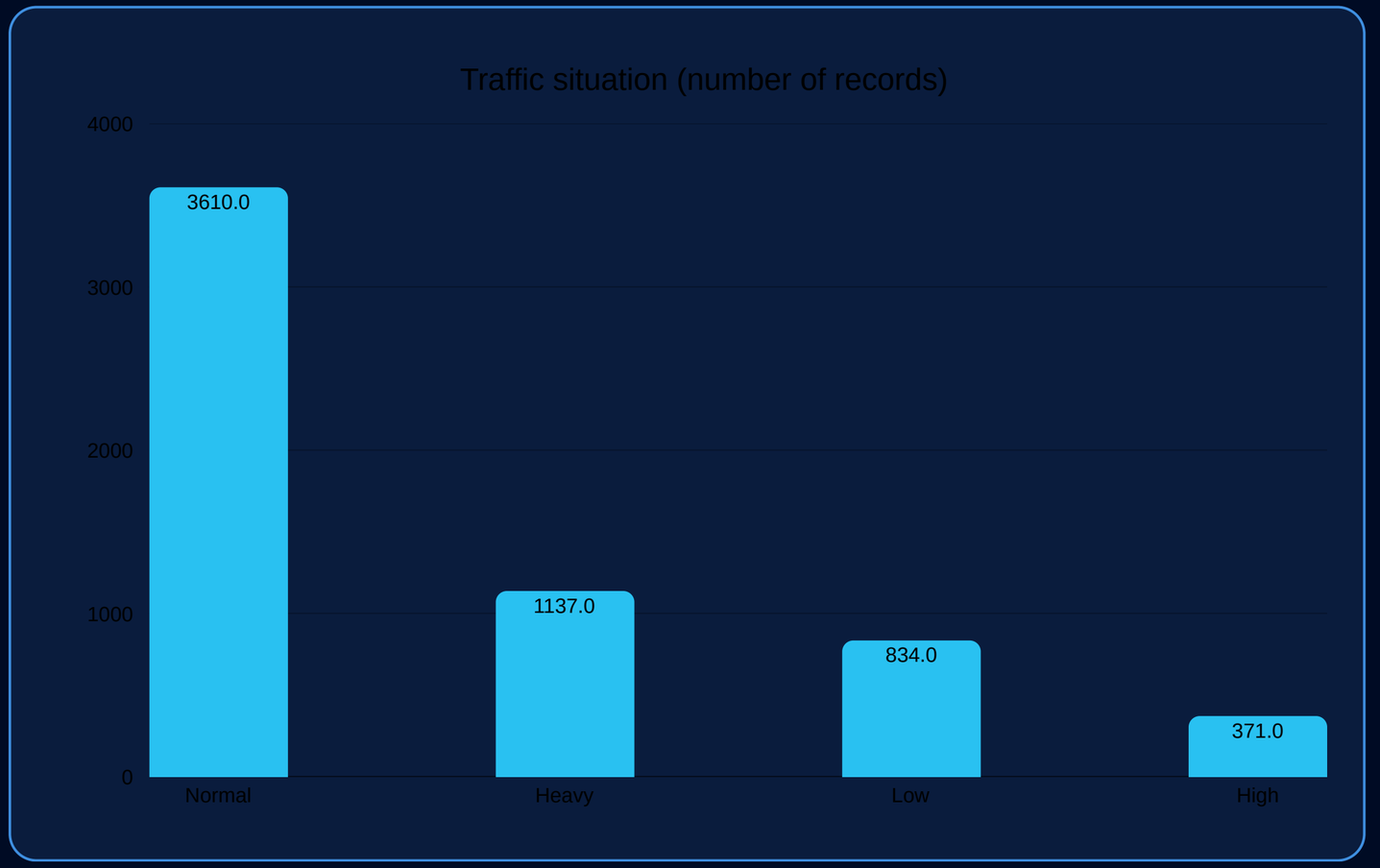
THE DATA BEHIND THE MODEL

5,952 records

62 days of data

15 min interval

Inputs: vehicle counts, weather, accidents, road closures, time and day, and past traffic patterns.



Normal traffic dominates and 'high' is rare, so class weights give the rare classes more influence.

FINAL MODELS: TWO TUNED MODELS

99.8%

classification accuracy on unseen days

17.8

mean absolute error, vehicles per 15 min

0.84

R^2 of the volume forecast (RMSE 23.5)

DECISION TREE CLASSIFIER

Labels the traffic situation: low, normal, high or heavy

- Class weights offset imbalance (3,610 normal vs 371 high)
- 99.8% accuracy vs 81.7% for Logistic Regression
- Tuned depth of 7 keeps it simple to read and explain

RANDOM FOREST REGRESSOR

Forecasts traffic volume (vehicles per 15 minutes)

- Logistics needs a number, not just a category
- Captures non-linear effects such as the 17:00 peak
- MAE 17.8 vs 21.7 for Linear Regression

TESTED LIKE A REAL LOGISTICS OPERATION

1. **Chronological split by day**
   4,608 rows (48 days) to train, 1,248 rows (13 days) to test

2. **TimeSeriesSplit, 5 folds**
   Trains on the past, tests on the future, so no leakage

3. **Class weights, no oversampling**
   Balanced weights lift the rare 'high' class; test days stay untouched

4. **Leakage check**
   Live counts add up to Total, so they never feed the volume forecast

17:00, road C closed Excluded by hard rule	PASS
17:00, accident on route A Heavy traffic, 67 min: not chosen	PASS
17:00, ring road B open and quiet Low traffic, 55 min: recommended	PASS
Forecast without live counts Accuracy drops to 61.6%	NOTE

Hard safety rule: if Road Closed = Yes, never recommend the route, whatever the model predicts.

TRAFFIC FOLLOWS A DAILY RHYTHM

0.84

R^2 between predicted and actual traffic volume on the unseen test days

TREND

Predictions track actual volume closely, with a consistent pattern across the three test days.

SEASONALITY

Volume falls overnight and peaks around 16:00. Traffic at the same time yesterday is a strong clue (autocorrelation 0.64).

WHERE IT STRUGGLES

Average error is about 23 vehicles in busy daytime hours (07:00 to 18:00) against about 12 overnight.

A SMALL NEURAL NETWORK TESTED



scikit-learn MLPRegressor with two ReLU hidden layers (64 and 32 neurons) on StandardScaler inputs. A linear output predicts vehicle volume.

TEST RESULT

MAE 19.6 and R^2 0.81 on unseen days: better than Linear Regression, behind the Random Forest.

EARLY STOPPING

Training stops when the validation score stops improving (up to 800 iterations), which limits overfitting.

WHY RANDOM FOREST WON

With about 4,600 training rows the forest was the most accurate model, so it is the one used for routing.

A VIRTUAL ASSISTANT FOR THE DRIVER



Which route should I take?

Route B, the ring road: about 55 minutes, low traffic, no accidents or closures.

Why not the highway?

Route A has heavy traffic and an accident, giving about 67 minutes.

Demo conversation from the chatbot prototype

Typed or spoken
Browser microphone for hands-free queries in a web chat. Replies can be read aloud (browser speech or pyttsx3).

Explains its choices
Gives reasons such as distance, expected traffic and current road conditions.

Real-time alerts
Reports accidents, closures, traffic and weather per route and recommends an alternative.

Reliable intent detection
Intent classifier scored 100% on 22 unseen phrases (117 training phrases, 11 intents).

The assistant supports route decisions. It does not guarantee safety.

TECHNIQUES MATCHED TO THE PROBLEM

Problem	Technique	Tool	Why it fits
Forecast traffic volume	Random Forest regression, tuned with RandomizedSearchCV	scikit-learn	MAE 17.8 vs 24.5 for the previous-interval baseline
Label the traffic state	Decision Tree with class weights, tuned	scikit-learn	99.8% accuracy on unseen days, easy to explain
Rank the routes	Adjusted time = free flow + congestion + accident and weather penalties; closed roads excluded	Python (routing.py)	Combines both models into one safest and fastest pick
Talk to the driver	TF-IDF + Logistic Regression intent classifier	Flask, Web Speech API	100% on 22 unseen phrases, with voice input

WHAT SAFEPATH DELIVERS

HARD RULE

Never recommend a closed road, whatever the models predict.

ADJUSTED TIME

Each route gets +15 min for an accident and +5 min per weather level; the lowest open route wins.

DRIVER DECIDES

The model advises. It does not drive the truck.

KEEP IT FRESH

Retrain weekly with TimeSeriesSplit as new traffic data arrives.

PREDICT. AVOID. DELIVER.

Fewer delays, safer drivers and goods, lower fuel use, and more reliable deliveries for local communities.

THANK YOU

QUESTIONS?

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